Assignment-1
Group 15
Design Document

We added packet dropping functionality in the e1000_xmit_frame() function which handles all outgoing packets.

Key Changes:

1. Added global variables:
    - blocked_ip - stores IP address to drop packets for
    - drop_rate - stores drop percentage (0-100)
2. Control packet detection:
    - Look for UDP packets with destination 100.100.100.100
    - Extract target IP (first 4 bytes of data)
    - Extract drop rate (5th byte of data)
    - Store these in global variables
3. Packet dropping:
    - Check if packet destination matches target
    - If drop_rate > 0, generate random number
    - If random < drop_rate, drop the packet
    - Use dev_kfree_skb_any() to free the buffer

Q1. Briefly explain your design and the key changes in e1000_main.c
Ans: Added global variables blocked_ip and drop_rate. In e1000_xmit_frame(), check for control packets to 100.100.100.100 and update these variables. For normal packets, if destination matches target, drop with given probability.

Q2. Transfer time without dropping packets?
350 ms

Q3. Retransmissions before sending packets?
1038

Q4. Retransmissions after sending packets (0% drop)?
1038

Q5. Transfer time with 1% packet drop?
689 ms

Q6. Retransmissions before sending packets (1% drop)?
1050

Q7. Retransmissions after sending packets (1% drop)?
1111

Q8. Transfer time with 5% packet drop?
6989ms

Q9. Retransmissions before sending packets (5% drop)?
1111

Q10. Retransmissions after sending packets (5% drop)?
1462

11. Transfer time with 10% packet drop?
30079 ms

Q12. Retransmissions before sending packets (10% drop)?
1462

Q13. Retransmissions after sending packets (10% drop)?
2341

Q14. Transfer time with 15% packet drop?
136362 ms

Q15. Retransmissions before sending packets (15% drop)?
2341

Q16. Retransmissions after sending packets (15% drop)?
3665

Q17. Does ping to Google work after dropping 100% packets?
Ans: No. All packets to Google's IP are dropped, so ping fails. (IP Used: 192.178.158.113)

Q18. Change in received packets after server buffer size change?
Ans: Yes, the number of received packets changed. With buf[64], the server received 100419 packets, while with buf[128], it received 50678 packets. This is because TCP is a byte-stream, so one recv() call does not necessarily correspond to one network packet. With a larger buffer, each recv() can receive more data, resulting in fewer recv() calls and hence a lower count.

Group Members:

1. Roshan Nepal 2024483
2. Lakshay Gupta 2024318
3. Arnav Gupta 2024108
4. Aishwary Soni 2024050
5. Punjani Rezaabbas Alihasain 2024445